

Effects of grass–endophyte symbiosis and herbivory on population demography across climatic and geographic gradients

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Abstract

Interactions between plants and fungi are ubiquitous in nature and have significant effects on plant fitness. However, the extent to which variation in fungal symbiosis affects plant demography across environmental gradients remains understudied. Using three cool-season grass species that host seed-transmitted fungal endophytes, we combined geographically-distributed field experiments across an aridity gradient and Bayesian modeling of demographic performance to test whether plant-fungi interactions affect host demography and, consequently, determine geographic range limits. We found that all vital rates decreased toward the species' range edges, indicating reduced demographic performance in these regions. This decline was more pronounced in individuals hosting endophytic symbionts, suggesting that plant-fungi interactions that are beneficial under benign conditions may become parasitic under stressful conditions. Our results highlight that [host resistance at range limits](#)¹ may be hindered by the presence of a mutualist.

¹*Tom wants us to think about this a bit more.*

Introduction

Plant-microbe symbioses are widespread and ecologically important. These interactions are famously context-dependent, where the direction and strength of the interaction outcome depends on the environment in which it occurs (Bronstein, 1994; Hoeksema and Bruna, 2015). Plant-microbe interactions that are beneficial under stressful conditions may become neutral or parasitic under benign conditions (Giauque et al., 2019). Under biotic stress (e.g., herbivory), endophyte symbiosis can benefit host plants by facilitating the production of secondary compounds that deter feeding or exert direct toxicity, thereby reducing insect growth, survival, and oviposition (Atala et al., 2022; Bastias et al., 2017; Vega, 2008). Similarly under abiotic stress (e.g., drought), symbionts can increase their host tolerance (Clay and Schardl, 2002). For example, mycorrhizal fungi form extensive networks that increase root surface area, helping plants absorb more water and nutrients such as phosphorus and nitrogen (Amijee et al., 1989). Symbiotic microbes also stimulate plants to produce antioxidant enzymes like superoxide dismutase and catalase that reduce oxidative damage caused by stresses such as drought, salinity, and heat (Ruiz-lazona et al., 1996). All these mechanisms help the host species better buffer against environmental stochasticity (Fowler et al., 2023).

Context dependence raises the hypothesis that plant-microbe interactions are likely to vary across environmental gradients, from range core to range edge, with significant implications for host range expansion at leading edges or resilience at trailing edges. Because range-edge environments are often more stressful due to factors such as extreme temperatures and limited soil nutrients (Hampe and Petit, 2005), the role of microbial symbionts may be especially important in these zones. If microbial symbiosis provides greater benefits under environmental stress, then symbionts could enhance the suitability of range-edge environments, potentially extending host range limits (Allsup et al., 2023; Rudgers et al., 2020). For instance, fungal endophytes improve the survival of *Bromus laevipes* populations in dry conditions, increasing their drought resistance at the range edge and thereby extending the species' geographic range (Afkhami et al., 2014; David et al., 2019). Even if the symbiont does not directly improve host survival, it may still enhance host population growth over time by increasing relative growth and reproduction, potentially offsetting the negative effects of lower survival rates (Yule et al., 2013). Conversely, if microbial symbiosis is costly for the host at range edge, symbionts could constrain host range (Bennett and Groten, 2022; Benning and Moeller, 2021a,b). For instance, symbiotic microbes, especially mycorrhizal fungi, require carbon subsidies from the plant (up to 20% of photosynthetically fixed carbon) (Soudzilovskaia et al., 2015; Thirkell et al., 2020). At range edges, photosynthesis may be limited, making the carbon cost harder to bear. Under such conditions, plants may reduce symbiosis, leading to host poor performance.

Despite growing interest in the role of symbiosis in host performance, few studies examined in natural conditions how abiotic and biotic stressors shape the effects of symbionts on host demography across geographic ranges (Louthan et al., 2015). Moreover, it remains unclear which stressor is most critical in eliciting the benefits of endophyte symbiosis, or how multiple stressors may interact to influence host demography. Herbivory is a pervasive biotic factor that can significantly affect host performance (Benning et al., 2019) and mediate the outcomes of symbiotic interactions across species' ranges. Herbivores reduce host fitness through tissue damage, defoliation, and altered resource allocation (Maron and Crone, 2006), but they may also shape the selective landscape in ways that favor symbiont persistence or transmission. For example, herbivory can enhance the fitness benefits of endophyte infection by triggering host defense pathways or promoting vertical transmission of the symbiont (Agrawal et al., 1999; Gundel et al., 2020). These effects may be especially pronounced at range margins, where plant populations often encounter intensified biotic pressures. If the ecological context of herbivory alters the net fitness effects of symbiosis, then interactions between herbivory and abiotic stress (e.g., drought) may play a critical role in determining population persistence and thereby species' range limits.

Working across a precipitation gradient in the south-central US, we investigated how endophyte symbiosis affects their host demography from core to edge. To do so, we studied the symbiotic association between three native cool-season grass species (*Agrostis hyemalis*, *Elymus virginicus* and *Poa autumnalis*) and their vertically transmitted fungal symbionts (*Epichloë* spp.). Our experiment was design to address the following:

1. Do plants with endophytes experience less damage from biotic stressors and maintain higher vital rates compared to plants without endophytes, when biotic stressors are present?
2. Under combined biotic and abiotic stress, do the mutualistic benefits of endophytes give host plants a performance advantage compared to those without endophytes?
3. Under combined biotic and abiotic stress, do the added physiological demands and resource costs of maintaining endophyte associations reduce overall plant performance, resulting in endophyte-infected plants performing worse than uninfected plants?

Materials and methods

Study system

Our study was conducted across seven sites along a geographic and climatic gradient spanning Louisiana and Texas in the United States (Fig. 1; Fig.S-1). At each site, we established eight plots (1.5 m × 1.5 m), with each plot containing 18 individuals of a single species. We used three cool-season perennial grasses native to woodland and prairie habitats of eastern North America (Shaw, 2011): *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnnalis* (subfamily Pooideae). They typically host seed-transmitted fungal endophytes. In summer–fall 2021, seeds were collected from naturally symbiotic populations (source populations) of the three focal host species (Fig. 1). These seeds were either heat-treated to eliminate symbionts or left untreated to generate symbiont-free (E^-) and symbiotic (E^+) plants from the same genetic lineages as the source populations.

Across all sites, each plot was randomly assigned one of four initial endophyte frequency treatments (80%, 60%, 40%, or 20%) and one of two herbivory treatments (herbivore exclusion or herbivore access). We ensured that all plots received comparable quantities of source plant material. This experimental design was informed by natural patterns of endophyte prevalence observed in Texas, where the proportion of symbiotic individuals has been reported as 86.55% in *A. hyemalis* (ranging from 77.16% to 93.5%) (Donald et al., 2021), 53% in *E. virginicus* (ranging from 10% to 100%) (Sneck et al., 2017), and 96% in *P. autumnnalis* (Rudgers et al., 2009). Additionally, fungal genotyping has confirmed the presence of biosynthetic pathways for secondary metabolites such as peramine, loline, and ergot alkaloids, which may enhance host resistance to drought and herbivory (Beaudry, 1951)². Full details of the study design are provided in the supplementary materials (Supplementary Method S.1).

Demographic data collection

We collected demographic data including survival, growth, and reproduction in May and June of 2023, 2024 and 2025, which coincided with the flowering season of *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnnalis*. For each individual, plant survival was recorded as a binary variable (alive or dead), and plant size was measured as the number of living tillers. We recorded the number of inflorescences per plant and counted the number of spikelets on up to three inflorescences from three reproducing plants. Spikelet counts were limited to three reproducing

²I am trying to justify why we choose the starting endophyte prevalence

tillers per plot due to the time-consuming nature of this measurement. We used the number of spikelets from these three tillers to estimate the average number of spikelets per plant.

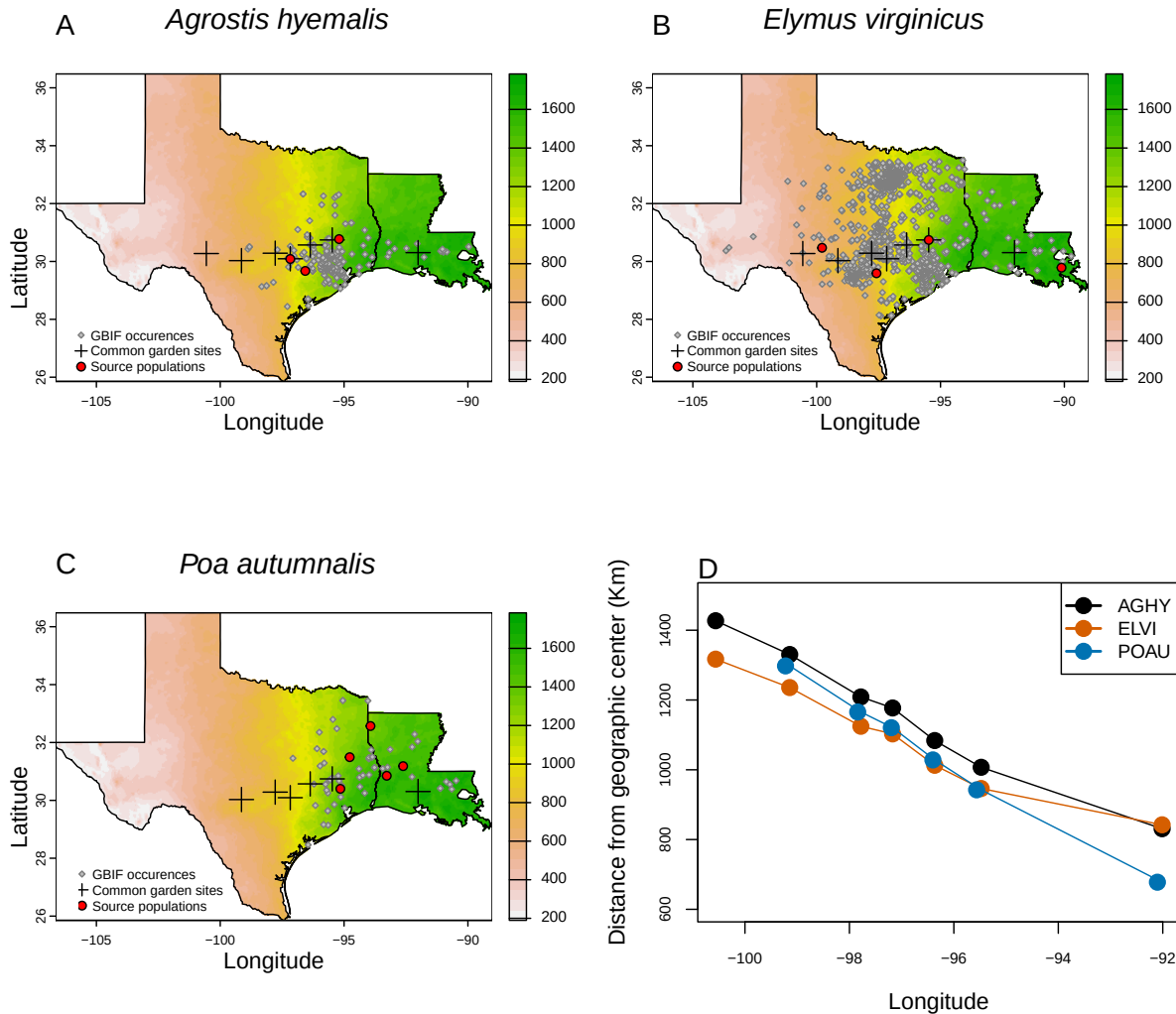


Figure 1: Distribution of common garden sites along the longitudinal aridity gradient in the central and southern Great Plains for: A) *Agrostis hyemalis* (AGHY), B) *Elymus virginicus* (ELVI), and C) *Poa autumnalis* (POAU). Red dots indicate the locations of source populations, while grey dots represent the Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area. D) Relationship between the distance from the range center and longitude for each species.

Models building and models selection

To assess how stress associated with aridity and herbivory affects plant–fungal interactions, we developed four candidate models for each vital rate (survival, growth, and fertility). Each vital rate was modeled using a grand mean intercept (β_0), slopes for variation in each covariate (β_1 ,

114 β_2 and β_3), and interaction terms between covariates (β_4 , β_5 and β_6). Each model included
 115 normally distributed random effects to account for site-to-site variation ($\phi \sim N(0, \sigma_{site})$),
 116 plot-to-plot variation ($\rho \sim N(0, \sigma_{plot})$), and source-to-source variation associated with the
 117 provenance of transplants used in the common garden ($\omega \sim N(0, \sigma_{source})$) (Eq. 1).

$$\begin{aligned}
 \text{Model 1: } \mu &= \beta_0^{sp} + \beta_P^{sp} P + \beta_{endo}^{sp} endo + \beta_{herb}^{sp} herb \\
 &\quad + \beta_{endo \times P}^{sp} endo \cdot P + \beta_{herb \times endo}^{sp} herb \cdot endo + \beta_{herb \times P}^{sp} herb \cdot P \\
 &\quad + \beta_{P^2}^{sp} P^2 + \beta_{endo \times P^2}^{sp} endo \cdot P^2 + \phi^{sp} + \omega^{sp} + \rho^{sp} \\
 \text{Model 2: } \mu &= \beta_0^{sp} + \beta_{GD}^{sp} GD + \beta_{endo}^{sp} endo + \beta_{herb}^{sp} herb \\
 &\quad + \beta_{endo \times GD}^{sp} endo \cdot GD + \beta_{herb \times endo}^{sp} herb \cdot endo + \beta_{herb \times GD}^{sp} herb \cdot GD \quad (1) \\
 &\quad + \phi^{sp} + \omega^{sp} + \rho^{sp} \\
 \text{Model 3: } \mu &= \beta_0^{sp} + \beta_{MD}^{sp} MD + \beta_{endo}^{sp} endo + \beta_{herb}^{sp} herb \\
 &\quad + \beta_{endo \times MD}^{sp} endo \cdot MD + \beta_{herb \times endo}^{sp} herb \cdot endo + \beta_{herb \times MD}^{sp} herb \cdot MD \\
 &\quad + \phi^{sp} + \omega^{sp} + \rho^{sp}
 \end{aligned}$$

119 where $sp \in \{1,2,3\}$ represents the species index, P denotes Precipitation, GD refers to distance
 120 from geographic center, and MD stands for mahalanobis distance, $endo$ indicates Endophyte
 121 status (absence or present), and $herb$ represents Herbivory (fenced and unfenced). The
 122 terms $\phi^{(sp)}$, $\omega^{(sp)}$, and $\rho^{(sp)}$ are plot-level, population-level, and site-level random effects,
 123 respectively, for species sp .

124 Survival was modeled using a Bernoulli distribution, growth with a Gaussian distribution,
 125 flowering with a negative binomial distribution, and fertility (number of spikelets) also with
 126 a negative binomial distribution. To assess the goodness-of-fit of the models, we performed
 127 posterior predictive checks (Berkhof et al., 2000; Gelman et al., 2000). All models effectively
 128 captured key aspects of the data, including means, standard deviations, and quantiles (Fig. ??,
 129 Fig. ??, Fig. ??, Fig. ??). To determine the best model for each vital rate, we compared the
 130 four models using leave-one-out cross-validation (LOOCV) (Vehtari et al., 2017). LOOCV
 131 combines validation and training by using one observation for testing and the remaining
 132 $n - 1$ observations for training. This process is repeated n times once for each observation
 133 resulting in n fitted models (Silva and Zanella, 2024). The test error estimate from LOOCV
 134 is obtained by averaging the prediction errors across all n models (Eq. 2).

$$CV_n = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

136 All models were implemented in R (R Core Team, 2023) using Stan via the RStan
137 interface (Stan Development Team, 2024).

Results

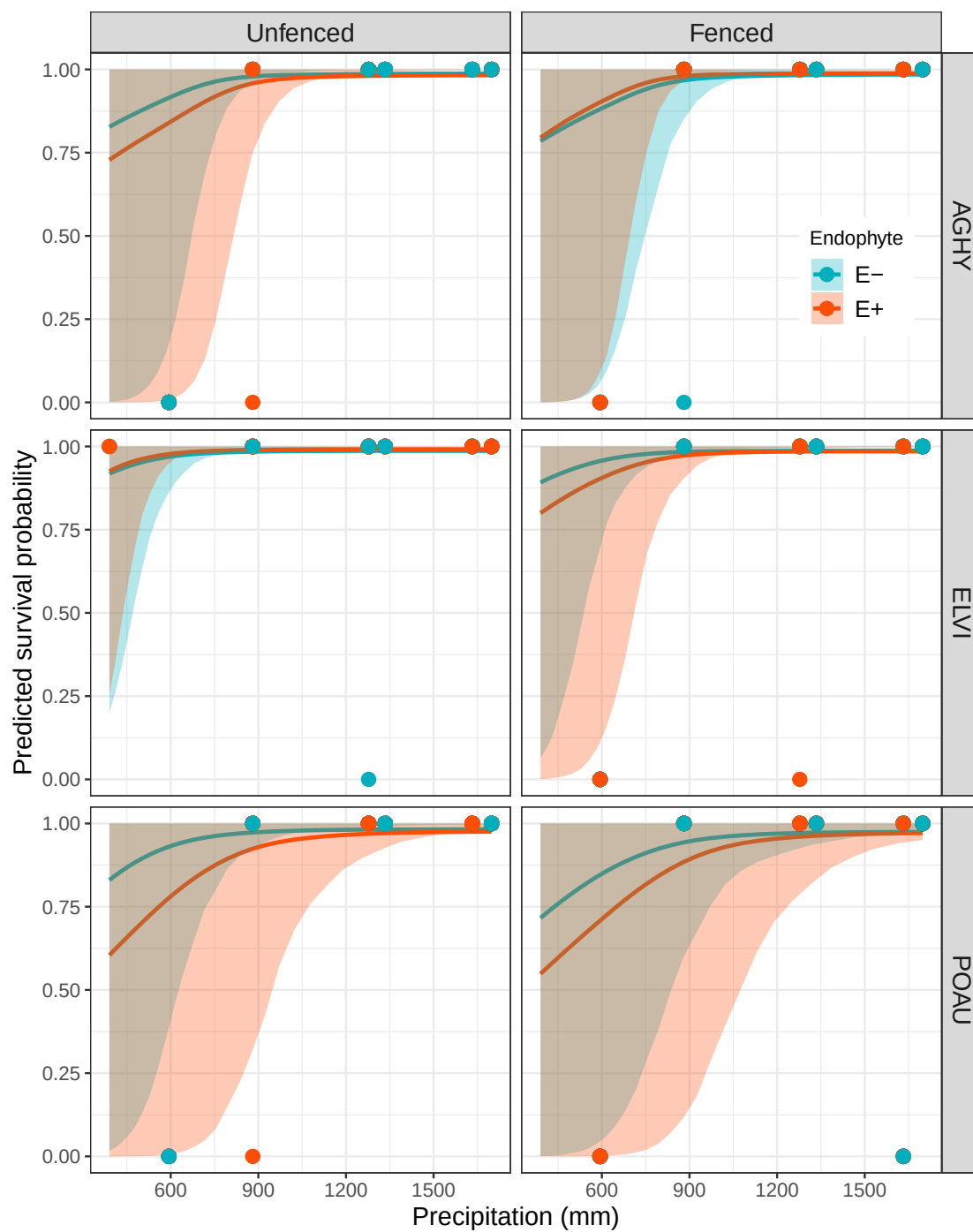


Figure 2: XXX

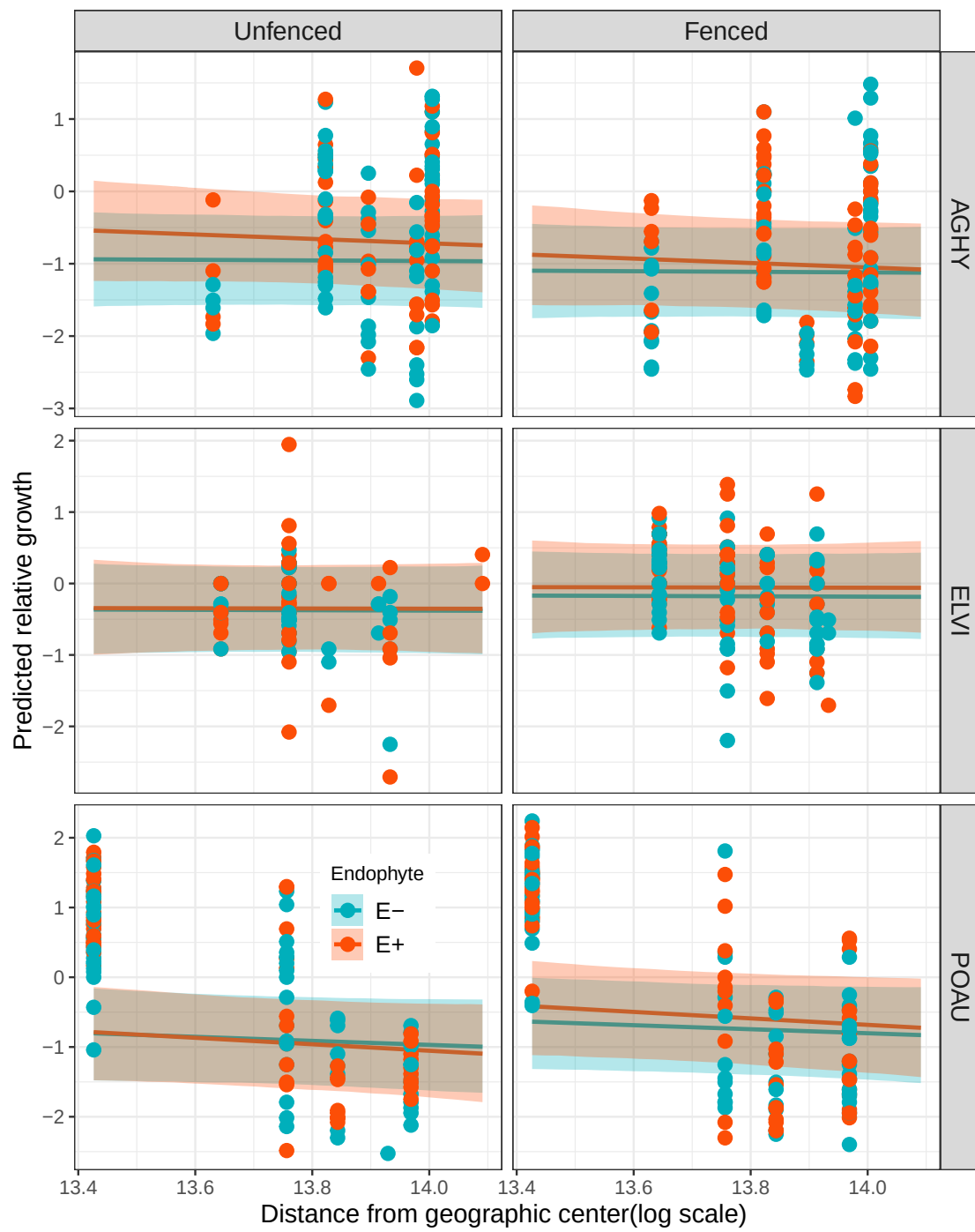


Figure 3: XXX

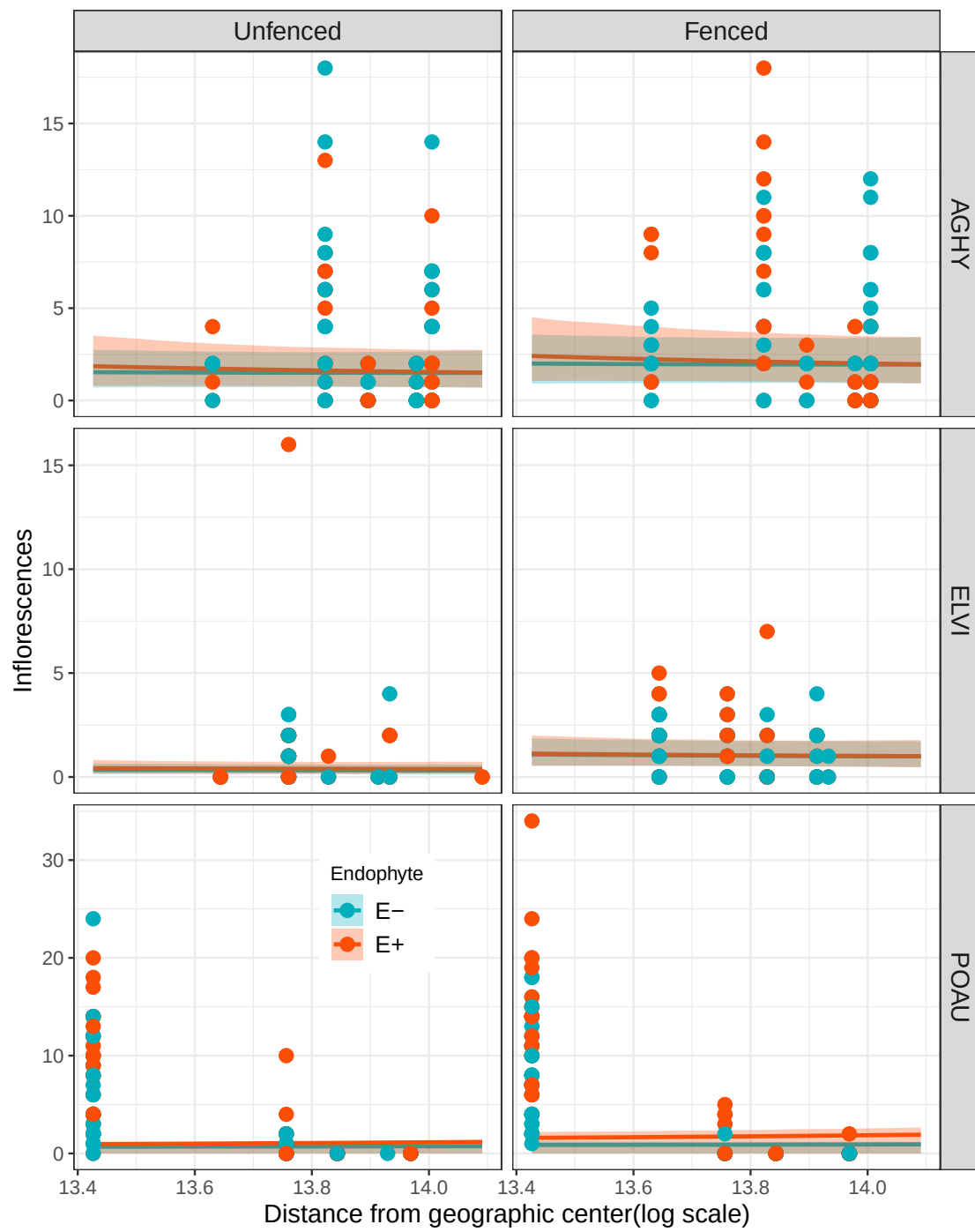


Figure 4: XXX

Acknowledgements

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References

- Afkhami, M. E., McIntyre, P. J., and Strauss, S. Y. (2014). Mutualist-mediated effects on species' range limits across large geographic scales. *Ecology letters*, 17(10):1265–1273.
- Agrawal, A. A., Laforsch, C., and Tollrian, R. (1999). Transgenerational induction of defences in animals and plants. *Nature*, 401(6748):60–63.
- Allsup, C. M., George, I., and Lankau, R. A. (2023). Shifting microbial communities can enhance tree tolerance to changing climates. *Science*, 380(6647):835–840.
- Amijee, F., Tinker, P., and Stribley, D. (1989). The development of endomycorrhizal root systems: Vii. a detailed study of effects of soil phosphorus on colonization. *New Phytologist*, 111(3):435–446.
- Atala, C., Acuña-Rodríguez, I. S., Torres-Díaz, C., and Molina-Montenegro, M. A. (2022). Fungal endophytes improve the performance of host plants but do not eliminate the growth/defence trade-off. *New Phytologist*, 235(2).
- Bastias, D. A., Martínez-Ghersa, M. A., Ballaré, C. L., and Gundel, P. E. (2017). Epichloë fungal endophytes and plant defenses: not just alkaloids. *Trends in Plant Science*, 22(11):939–948.
- Beaudry, J.-R. (1951). Seed development following the mating elymus virginicus l. x agropyron repens (l.) beauv. *Genetics*, 36(2):109.
- Bennett, A. E. and Groten, K. (2022). The costs and benefits of plant–arbuscular mycorrhizal fungal interactions. *Annual Review of Plant Biology*, 73(1):649–672.
- Benning, J. W., Eckhart, V. M., Geber, M. A., and Moeller, D. A. (2019). Biotic interactions contribute to the geographic range limit of an annual plant: herbivory and phenology mediate fitness beyond a range margin. *The American Naturalist*, 193(6):786–797.
- Benning, J. W. and Moeller, D. A. (2021a). Microbes, mutualism, and range margins: testing the fitness consequences of soil microbial communities across and beyond a native plant's range. *New Phytologist*, 229(5):2886–2900.
- Benning, J. W. and Moeller, D. A. (2021b). Plant–soil interactions limit lifetime fitness outside a native plant's geographic range margin. *Ecology*, 102(3):e03254.
- Berkhof, J., Van Mechelen, I., and Hoijsink, H. (2000). Posterior predictive checks: Principles and discussion. *Computational Statistics*, 15:337–354.

- Bronstein, J. L. (1994). Conditional outcomes in mutualistic interactions. *Trends in ecology & evolution*, 9(6):214–217.
- Church, G. L. (1958). Artificial hybrids of *elymus virginicus* with *e. canadensis*, *interruptus*, *riparius*, and *wiegandii*. *American Journal of Botany*, pages 410–417.
- Clay, K. and Schardl, C. (2002). Evolutionary origins and ecological consequences of endophyte symbiosis with grasses. *the american naturalist*, 160(S4):S99–S127.
- David, A. S., Quintana-Ascencio, P. F., Menges, E. S., Thapa-Magar, K. B., Afkhami, M. E., and Searcy, C. A. (2019). Soil microbiomes underlie population persistence of an endangered plant species. *The American Naturalist*, 194(4):488–494.
- Donald, M. L., Bohner, T. F., Kolis, K. M., Shadow, R. A., Rudgers, J. A., and Miller, T. E. (2021). Context-dependent variability in the population prevalence and individual fitness effects of plant–fungal symbiosis. *Journal of Ecology*, 109(2):847–859.
- Fowler, J. C., Donald, M. L., Bronstein, J. L., and Miller, T. E. (2023). The geographic footprint of mutualism: How mutualists influence species’ range limits. *Ecological Monographs*, 93(1):e1558.
- Gelman, A., Goegebeur, Y., Tuerlinckx, F., and Van Mechelen, I. (2000). Diagnostic checks for discrete data regression models using posterior predictive simulations. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 49(2):247–268.
- Giauque, H., Connor, E. W., and Hawkes, C. V. (2019). Endophyte traits relevant to stress tolerance, resource use and habitat of origin predict effects on host plants. *New Phytologist*, 221(4):2239–2249.
- Gundel, P. E., Sun, P., Charlton, N. D., Young, C. A., Miller, T. E., and Rudgers, J. A. (2020). Simulated folivory increases vertical transmission of fungal endophytes that deter herbivores and alter tolerance to herbivory in *poa autumnalis*. *Annals of Botany*, 125(6):981–991.
- Hampe, A. and Petit, R. J. (2005). Conserving biodiversity under climate change: the rear edge matters. *Ecology letters*, 8(5):461–467.
- Hoeksema, J. D. and Bruna, E. M. (2015). Context-dependent outcomes of mutualistic interactions. *Mutualism*, 10:181–202.
- Louthan, A. M., Doak, D. F., and Angert, A. L. (2015). Where and when do species interactions set range limits? *Trends in ecology & evolution*, 30(12):780–792.

- Maron, J. L. and Crone, E. (2006). Herbivory: effects on plant abundance, distribution and population growth. *Proceedings of the Royal Society B: Biological Sciences*, 273(1601):2575–2584.
- R Core Team (2023). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.
- Rudgers, J. A., Afkhami, M. E., Bell-Dereske, L., Chung, Y. A., Crawford, K. M., Kivlin, S. N., Mann, M. A., and Nuñez, M. A. (2020). Climate disruption of plant-microbe interactions. *Annual review of ecology, evolution, and systematics*, 51:561–586.
- Rudgers, J. A., Afkhami, M. E., Rúa, M. A., Davitt, A. J., Hammer, S., and Huguet, V. M. (2009). A fungus among us: broad patterns of endophyte distribution in the grasses. *Ecology*, 90(6):1531–1539.
- Ruiz-lazona, J. M., Azcón, R., and Palma, J. (1996). Superoxide dismutase activity in arbuscular mycorrhizal lactuca sativa plants subjected to drought stress. *New Phytologist*, 134(2):327–333.
- Shaw, R. B. (2011). *Guide to Texas grasses*. Texas A&M University Press.
- Silva, L. A. and Zanella, G. (2024). Robust leave-one-out cross-validation for high-dimensional bayesian models. *Journal of the American Statistical Association*, 119(547):2369–2381.
- Sneck, M. E., Rudgers, J. A., Young, C. A., and Miller, T. E. (2017). Variation in the prevalence and transmission of heritable symbionts across host populations in heterogeneous environments. *Microbial Ecology*, 74:640–653.
- Soudzilovskaia, N. A., van der Heijden, M. G., Cornelissen, J. H., Makarov, M. I., Onipchenko, V. G., Maslov, M. N., Akhmetzhanova, A. A., and van Bodegom, P. M. (2015). Quantitative assessment of the differential impacts of arbuscular and ectomycorrhiza on soil carbon cycling. *New Phytologist*, 208(1):280–293.
- Stan Development Team (2024). RStan: the R interface to Stan. R package version 2.32.6.
- Thirkell, T. J., Pastok, D., and Field, K. J. (2020). Carbon for nutrient exchange between arbuscular mycorrhizal fungi and wheat varies according to cultivar and changes in atmospheric carbon dioxide concentration. *Global change biology*, 26(3):1725–1738.
- Vega, F. E. (2008). Insect pathology and fungal endophytes. *Journal of invertebrate pathology*, 98(3):277–279.
- Vehtari, A., Gelman, A., and Gabry, J. (2017). Practical bayesian model evaluation using leave-one-out cross-validation and waic. *Statistics and computing*, 27:1413–1432.

²³¹ Yule, K. M., Miller, T. E., and Rudgers, J. A. (2013). Costs, benefits, and loss of vertically
²³² transmitted symbionts affect host population dynamics. *Oikos*, 122(10):1512–1520.

Supporting Information

S.1 Supporting methods

Study design

Experimental Design To understand the demographic effects of endophyte symbiosis from core to edge of the host range, we established common gardens at 7 sites across the geographic range of *Elymus virginicus* (Fig. S-1). Experimental sites spanned an aridity gradient (temperature gradient). Common gardens were established in 8 plots per site. Plots were 1.5m * 1.5m and the area was tilled of existing vegetation to control for native plant competition. Plots were also selected in shaded areas under tree canopy or near shrubs to mimic the natural environmental of the species. In each plot, we planted 15 individuals of *E. virginicus* approximately 15 cm deep in an evenly spaced 4*4 grid pattern, with positions randomly assigned. For each plot, we randomly assigned a starting endophyte frequency (80%, 60%, 40%, 20%)³ and herbivory treatment (herbivores exclusion and herbivores accessibility). We ensured that all plots had comparable quantities of source populations. After establishing the plots, we watered the plants and recorded initial tiller counts, flowering status and plot position, endophyte status, source population of each individual plant. For herbivory exclusion plots, we enclosed them with 1.2m tall mesh fencing to prevent browsing by vertebrate herbivores and sprayed the plots with insecticide. For herbivores accessibility plots (control treatment), we half enclosed the plots with the mesh netting.

Source populations and Identification of individual endophyte status Plants used in the common garden experiment were derived from natural populations throughout the native range in the south-central US. At each of these natural populations we collected seeds. Some of the seeds of *E. virginicus* were heat treated to produce endophyte negative plants (E^-). To do so, we placed these seeds in a drying oven set at 60°C for approximately five days (120 hours). While this method eliminates the endophytes from all individuals, it does not affect seed viability. All seeds (both heat-treated and non-heat-treated) were planted in the Rice University greenhouse. Seedlings were regularly fertilized every two weeks. The seedlings were then vegetatively propagated to produce enough individuals for your experiment (N = 840). Before planting in the field, we confirmed the endophyte status (E^+ or E^-) of all seedlings using either microscopy or an immunoblot assay. This was necessary due to the varying success of the heat treatment and differences in the prevalence of endophytes between

³Do we need a schematic of one replicate of the experimental design?

the natural populations. Leaf tissues were stained with aniline blue lactic acid and viewed under a compound microscope at 200x-400x to identify fungal hyphae. The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of *Epichloë* spp. and chromagen to visually indicate presence or absence. Both methods yield similar detection rates.

S.2 Supporting Tables

Table S-1: Populations and Coordinates

Population	Location	Coordinates	NE+ / NE-
Sam Houston State University Field Station	Huntsville, TX	30°44'30.5"N, 95°28'28.2"W	NE+ = 39, NE- = 50
Texas Tech University	Junction, TX	30°28'18.2"N, 99°47'01.7"W	NE+ = 49, NE- = 83
Palmetto State Park	Palmetto, TX	29.591623, -97.584781	NE+ = 250, NE- = 192
Jean Lafitte National Historical Park and Preserve	Jean Lafitte, LA	29.785105, -90.114933	NE+ = 83, NE- = 94

Table S-2: Model comparison by vital rate using ELPD (Expected Log Pointwise Density) differences. The values in the **elpd_diff** column represent the difference in the model's ELPD relative to the best model for each vital rate, with positive values indicating a worse model fit. The **se_diff** column shows the standard error of the ELPD difference. Best models for each vital rate are bolded. Model selection is based on the ELPD difference and associated standard error, with smaller ELPD differences and lower standard errors indicating better model fit.

Vital rate	Model	elpd_diff	se_diff
Survival	model1	0.0	0.0
	model2	-1.0	1.5
	model3	-1.0	1.5
Growth	model3	0.0	0.0
	model2	-1.3	2.0
	model1	-2.1	2.2
Flowering	model2	0.0	0.0
	model3	0.0	3.2
	model1	-1.3	3.9

S.3 Supporting Figures

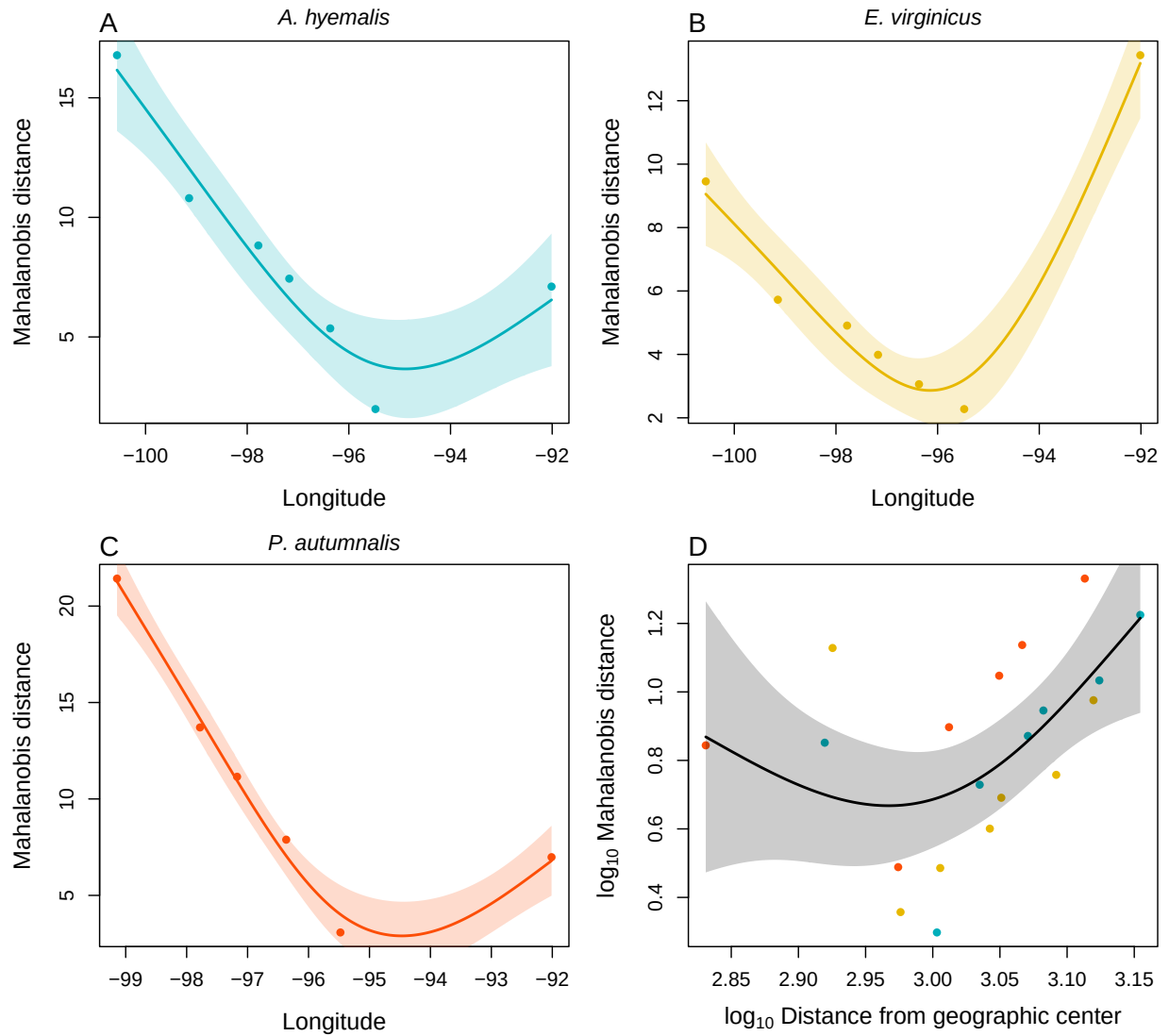


Figure S-1: (A-C) Relationship between distance from geographic center and longitude for the focal species (*Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*). A generalized additive model (GAM) with a smooth term was used to capture the non-linear trends in these relationships for each species. The solid lines represent the fitted GAM with shaded regions showing the 95% confidence interval. D) displays the relationship between the distance from geographic center and the distance from niche centroid for all species also fitted with a GAM on a log scale. The p-value for the smooth term in the GAM model is $p=0.0166^{**}$, indicating a significant relationship between log-transformed geographic distance and Mahalanobis distance.

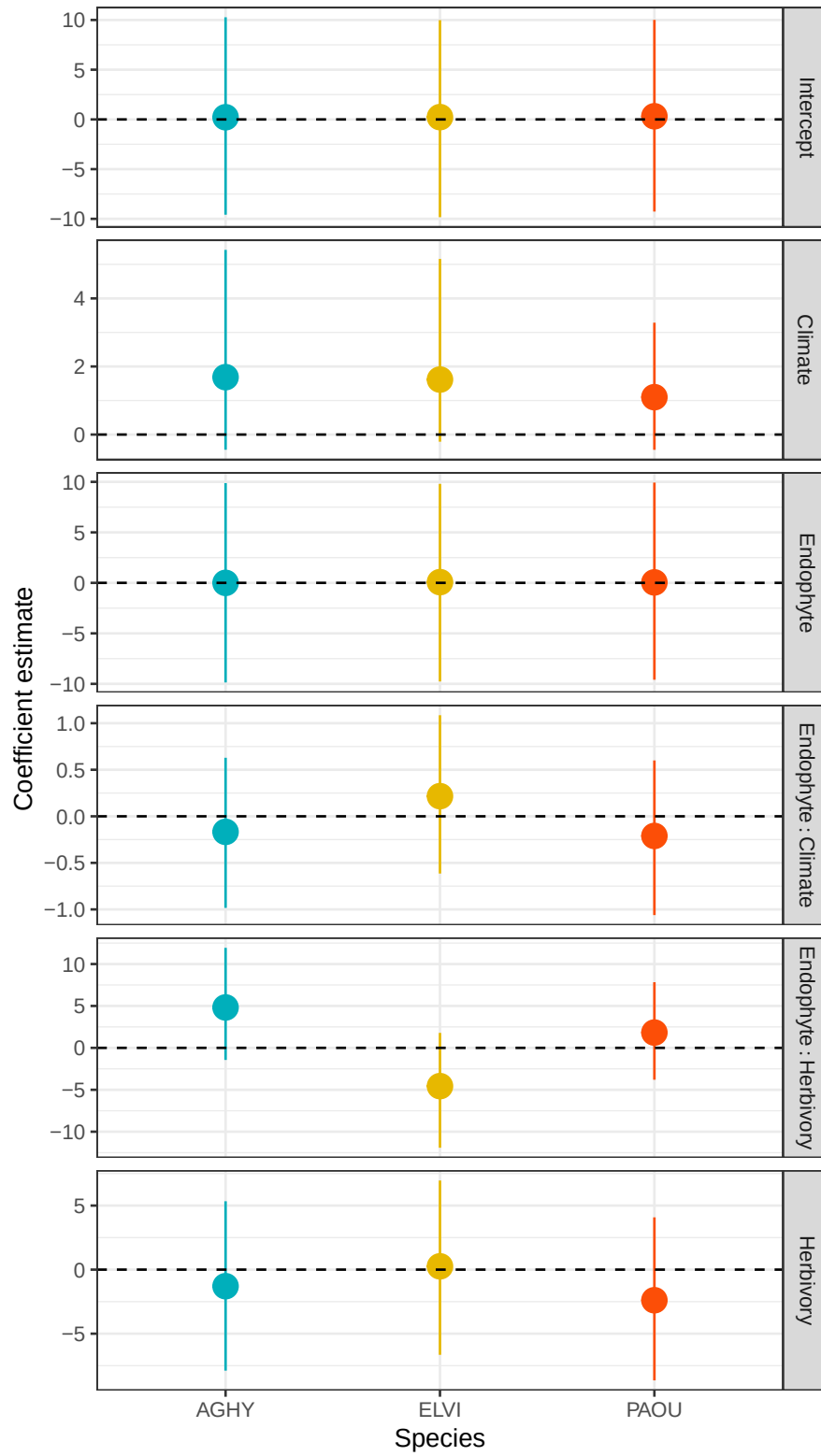


Figure S-2: Coefficient estimates from Bayesian models linking survival and geographic distance. Points represent posterior means and lines indicate 95% credible intervals. Species abbreviations: *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (PAOU). Horizontal dashed lines denote zero effect.

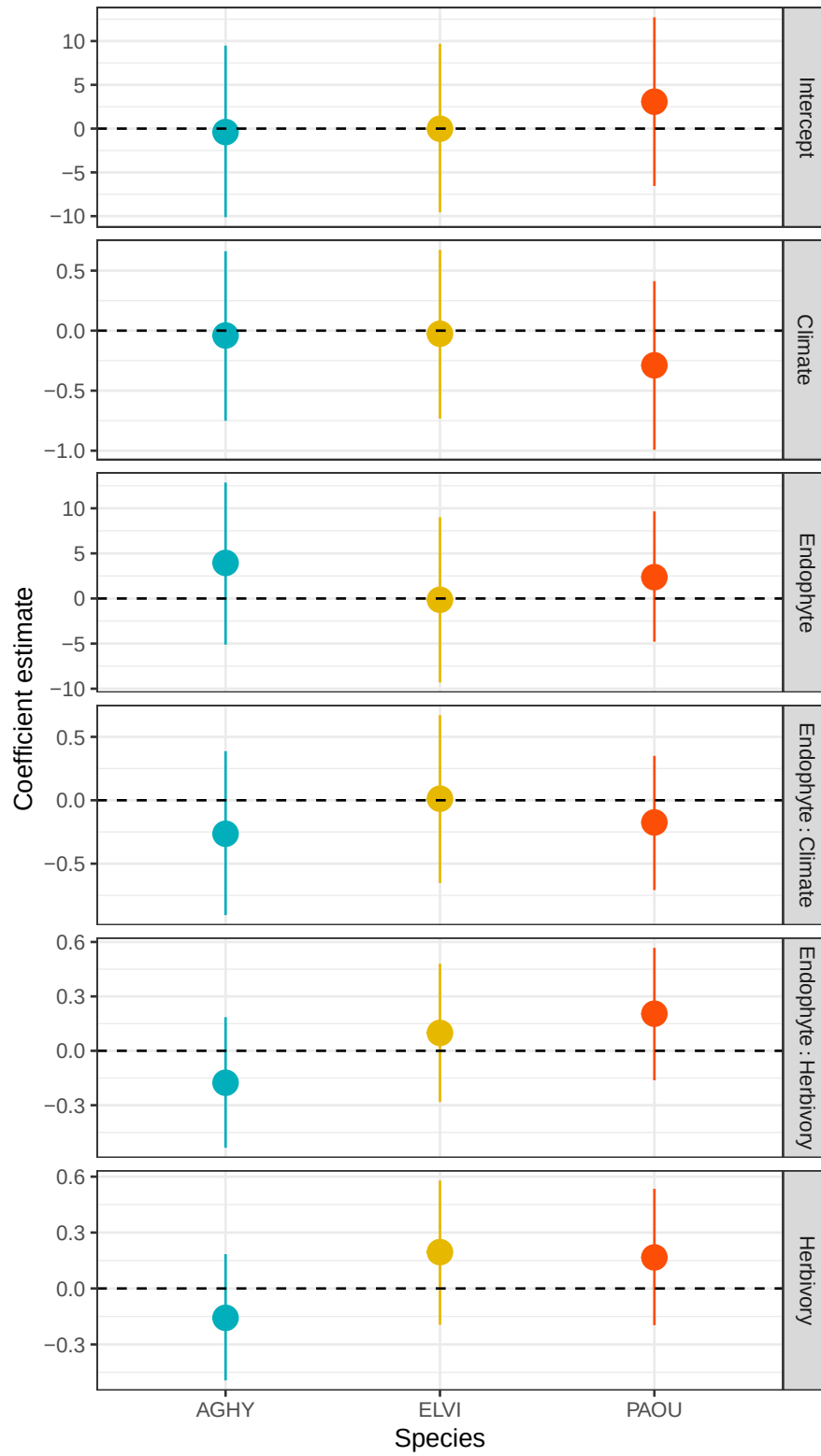


Figure S-3: Coefficient estimates from Bayesian models linking growth and geographic distance. Points represent posterior means and lines indicate 95% credible intervals. Species abbreviations: *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (PAOU). Horizontal dashed lines denote zero effect.

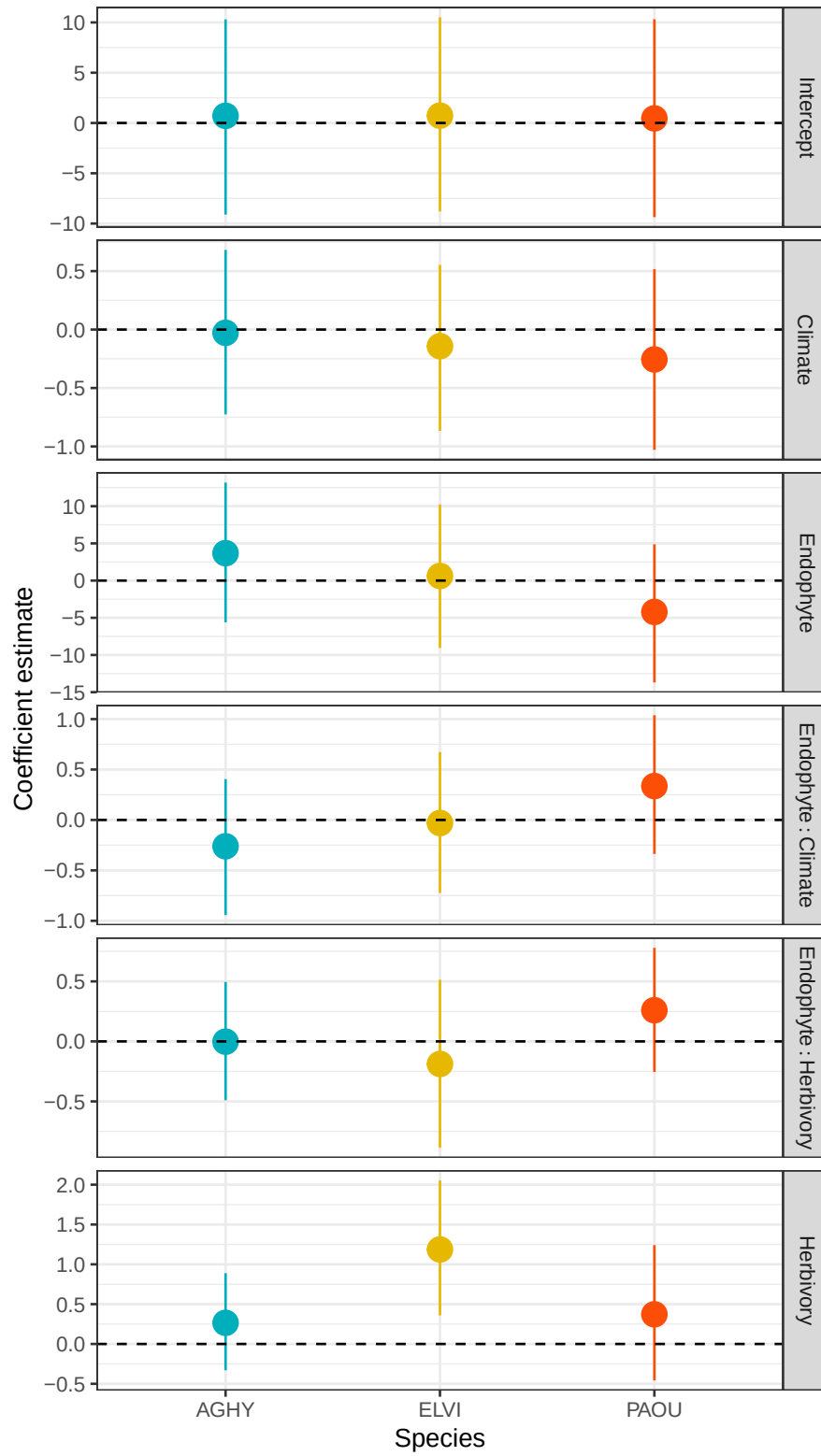


Figure S-4: Coefficient estimates from Bayesian models linking flowering and geographic distance. Points represent posterior means and lines indicate 95% credible intervals. Species abbreviations: *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (PAOU). Horizontal dashed lines denote zero effect.